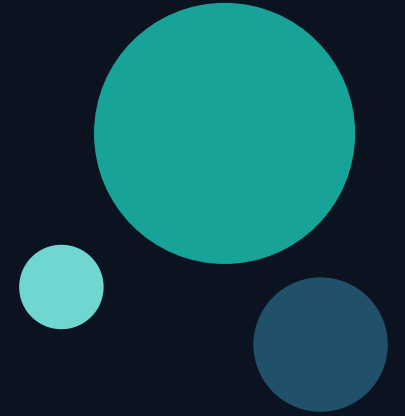


MariaDB Administration

POJ Putrajaya

Two-hour instructor-led demonstration

From server structure to safe daily operations



2 HOURS / LIVE DEMO

What we will do in two hours

- 01 Understand the MariaDB structure
- 02 Inspect the running service and data directory
- 03 Demonstrate accounts, grants, and least privilege
- 04 Use transactions, indexes, and EXPLAIN
- 05 Back up, restore, and verify persistence
- 06 Perform first-line operational checks

AUDIENCE

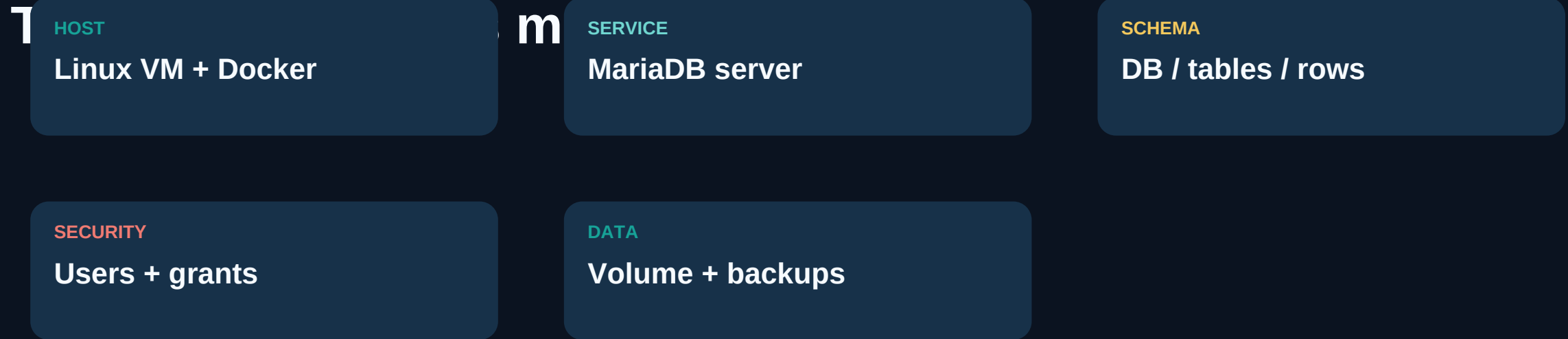
POJ IT personnel

STYLE

Instructor-led

OUTCOME

Operational confidence



DATA Named volume, backups, restore tests

The POJ demonstration topology

VM

CKAD golden image with
Docker already installed

CONTAINER

poj-mariadb

IMAGE

mariadb:11.4

VOLUME

poj-mariadb-data

ENDPOINT

127.0.0.1:3306

The service is local to the VM; the demo is not an internet-facing database.

First response: is the service healthy?

- Container state: docker ps
- Readiness: mariadb-admin ping
- Identity: VERSION(), @@hostname, @@datadir
- Workload: SHOW FULL PROCESSLIST
- Signals: logs, connections, threads, resource usage

ADMINISTRATOR'S LENS

Health is a set of signals, not a single green light.

Use the Educates command then pause for discussion.

Accounts and least privilege

- Identity includes username AND host
- Root is for administration, not applications
- Application account: access to poj_demo
- Reporting account: SELECT only
- Review with SHOW GRANTS
- Test the boundary: a denied INSERT is useful evidence

ADMINISTRATOR'S LENS

Grant the smallest practical scope and review it regularly.

Use the Educates command then pause for discussion.

Safe change: transaction first

- START TRANSACTION
- Read the current state
- Make a controlled change
- COMMIT when correct
- ROLLBACK when the outcome is wrong
- Use EXPLAIN before changing a slow query

ADMINISTRATOR'S LENS

A transaction is a unit of work. An index is a trade-off, not a free performance switch.

Use the Educates command
then pause for discussion.

Persistence is separate from the container

- Container writable layer: disposable
- Named volume: database files survive container replacement
- Logical dump: portable SQL representation
- Backup policy: retention, encryption, off-host copy
- Restore test: proof that the backup is useful

ADMINISTRATOR'S LENS

Never confuse “the container restarted” with “the database is protected.”

Use the Educates command
then pause for discussion.

Backup and restore demonstration

- Create: mariadb-dump --single-transaction
- Inspect: schema, tables, inserts, routines, triggers
- Remove a demo object
- Restore into the database
- Verify the object and rows are back

ADMINISTRATOR'S LENS

A successful backup command is not the same as a tested recovery procedure.

Use the Educates command then pause for discussion.

Daily operational checks

- Availability: mariadb-admin ping
- Sessions: SHOW FULL PROCESSLIST
- Connections: Threads_connected / Connections
- Storage: information_schema.tables
- Logs: docker logs --tail 30
- Safe restart: restart service, keep volume

ADMINISTRATOR'S LENS

Every action should have an owner, a change record, and a rollback path.

Use the Educates command then pause for discussion.

Instructor click-and-run runbook

- 1 Welcome and predict the result
- 2 Inspect container, server, schemas, and data
- 3 Create reporting account and prove access boundary
- 4 Commit/rollback and inspect an index
- 5 Dump, remove, restore, verify
- 6 Run health and cleanup checks

ADMINISTRATOR'S LENS

Pause after each command and connect the result to a production responsibility.

Use the Educates command then pause for discussion.

The POJ administration mindset

- Know where the service runs
- Know who can do what
- Know where data lives
- Know how to recover it
- Know which signals show trouble
- Replace workshop credentials before any real deployment

ADMINISTRATOR'S LENS

MariaDB administration is disciplined operation: access, change, protection, observation, recovery.

Use the Educates command
then pause for discussion.